

Movie Genre Classification

This repository contains my solution for **Task 1: Movie Genre Classification**.

Task Objective

Build a machine learning model that predicts the **genre of a movie** based on its **plot summary / description**.

Dataset

[Genre Classification Dataset \(IMDb\) — Kaggle](#)

Format:




```
train_data.txt      → ID ::: TITLE ::: GENRE ::: DESCRIPTION
test_data.txt       → ID ::: TITLE ::: DESCRIPTION
test_data_solution.txt → ID ::: TITLE ::: GENRE ::: DESCRIPTION
```

Approach

Text Cleaning — lowercasing, removing URLs, punctuation, and numbers.

Feature Extraction — TF-IDF Vectorization (unigrams + bigrams, 20,000 features).

Model Training — trained and compared three classifiers:

Multinomial Naive Bayes

Logistic Regression

Linear SVM (LinearSVC)

Evaluation — accuracy score, classification report, and confusion matrix.

Prediction — best-performing model saved and used to predict genres on held-out data.

Results

Model	Validation Accuracy
Naive Bayes	<i>(printed when you run the script)</i>
Logistic Regression	<i>(printed when you run the script)</i>
Linear SVM	<i>(printed when you run the script)</i>

Running the script generates `model_comparison.png` and `confusion_matrix.png` in the project root with the actual numbers for your run.

Tech Stack

Python

Pandas, NumPy

Scikit-learn (TF-IDF, Naive Bayes, Logistic Regression, SVM)

Matplotlib, Seaborn



How to Run

Clone the repo and move into it:



bash

```
git clone https://github.com/<your-username>/movie-genre-classification.git
cd movie-genre-classification
```

(Recommended) Create a virtual environment:



bash

```
python -m venv venv
# Windows
venv\Scripts\activate
# Mac/Linux
source venv/bin/activate
```

Install dependencies:



bash

```
pip install -r requirements.txt
```

Get the dataset and place these files directly inside `data/`:

```
data/train_data.txt
```

```
data/test_data_solution.txt
```

Either download manually from Kaggle, or use the Kaggle CLI (requires a `kaggle.json` API token in

```
~/.kaggle/)
```



bash

```
kaggle datasets download -d hijest/genre-classification-dataset-imdb -p data --unzip
```

Run the script:



bash

```
python src/movie_genre_classification.py
```

This will:

Load and clean the plot summaries (lowercase, strip URLs/punctuation, remove stopwords)

Vectorize text with TF-IDF (unigrams + bigrams, 20k features)

Train and evaluate Naive Bayes, Logistic Regression, and Linear SVM

Print accuracy and a classification report for each model

Save `model_comparison.png` and `confusion_matrix.png` to the project root

Save `test_predictions.csv` with true vs. predicted genres on the held-out split

Save the best-performing model + TF-IDF vectorizer to `models/`

Repository Structure

```

movie-genre-classification/
├── data/                # dataset files (not committed, see .gitignore)
│   ├── train_data.txt
│   └── test_data_solution.txt
├── models/              # saved model + vectorizer (not committed)
│   ├── genre_model.pkl
│   └── tfidf_vectorizer.pkl
├── src/
│   └── movie_genre_classification.py # main script
├── model_comparison.png  # generated after running the script
├── confusion_matrix.png  # generated after running the script
├── test_predictions.csv   # generated after running the script
├── requirements.txt
├── .gitignore
├── LICENSE
└── README.md
```

Predicting on new text

```

python

from src.movie_genre_classification import predict_genre

genre = predict_genre(
    "A group of astronauts travel through a wormhole in search of a new home for humanity.",
    "models/genre_model.pkl",
    "models/tfidf_vectorizer.pkl"
)

print(genre)
```

License

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